

Institute of Actuaries of India

Subject SA6 – Investment

May 2014 Examinations

INDICATIVE SOLUTIONS

Solution 1 :

1)

i.

Cost of equity

Geometric average dividend growth rate = $(21.8/19.38)^{(1/4)} - 1 = 0.0298$ or 3%

Using the dividend growth model, $k_e = 0.03 + ((21.8 \times 1.03)/250) = 0.03 + 0.09 = 12\%$

Market values of equity and debt

Market value of equity = $V_e = 1\text{m} \times 250 = \text{Rs } 250$ million

Market value of bonds = $V_d = 60\text{m} \times (104/100) = \text{Rs } 62.4$ million

Total market value of ABC Co = $V_e + V_d = 250 + 62.4 = \text{Rs } 312.4$ million

Current WACC calculation

The current after-tax cost of debt is 7%

$\text{WACC} = ((k_e \times V_e) + (k_d(1 - T) \times V_d)/(V_e + V_d)) = ((12 \times 250\text{m}) + (7 \times 62.4\text{m}))/312.4\text{m} = 11\%$

The weighted average after-tax cost of capital before the new issue of bonds is 11%

After-tax cost of debt of new bond issue

After-tax interest rate = $8 \times (1 - 0.3) = 5.6\%$ per year

Using linear interpolation:

Year	Cash flow	PV (Rs)	5% Discount	PV (Rs)	6% Discount	PV (Rs)
0	Market value	(100)	1.000	(100.00)	1.000	(100.00)
1–10	Interest	5.6	7.722	43.24	7.360	41.22
10	Redemption	105	0.614	64.47	0.558	58.59
				<u>7.71</u>		<u>(0.19)</u>

After-tax cost of debt = $5 + [((6 - 5) \times 7.71)/(7.71 + 0.19)] = 5 + 0.98 = 5.98\%$ or 6%

Revised calculation

The market value of the new issue of bonds is Rs 40 million

The total market value of ABC Co increases to $312.4 + 40 = \text{Rs } 352.4$ million

Revised WACC = $((12 \times 250\text{m}) + (7 \times 62.4\text{m}) + (6 \times 40))/352.4\text{m} = 10.4\%$

[8]

- ii. There is certainly a relationship between the weighted average cost of capital (WACC) and the market value of the company, since the market value can be expressed as the present value of future corporate cash flows, discounted by the WACC.

Marginal and average cost of debt

As for decreasing the WACC by issuing traded bonds, if the marginal cost of capital, in this case the cost of debt of the new bond issue, is less than the weighted average cost of capital (WACC), it would

seem logical to expect the WACC to decrease. However, as noted in an earlier discussion, increasing gearing will increase financial risk and may lead to an increase in the cost of equity, offsetting the effect of the cheaper debt.

Traditional view of capital structure

In the traditional view of capital structure, there is a non-linear relationship between the cost of equity and financial risk, as measured by gearing. Equity investors are indifferent to the addition of small amounts of debt, so as a company gears up by replacing expensive equity with cheaper debt, the WACC initially decreases. Debt is cheaper than equity because of the relative positions of the two sources of finance in the creditor hierarchy (the traditional view of capital structure ignores taxation). As equity investors start to respond to increasing financial risk, however, the cost of equity begins to increase until a point is reached where WACC ceases to fall. This corresponds to an optimal capital structure, since at this point WACC is at a minimum and hence the market value of the company is at a maximum. After this point, the WACC starts to increase as the company continues to gear up, rising more quickly at very high levels of gearing due to the appearance of bankruptcy risk. Under the traditional view the finance director might be correct in his belief that issuing debt will decrease WACC, depending on the position of the company relative to its optimal capital structure.

Imperfect market

In reality, it was noted that companies do not gear up as much as possible because of the dangers of high gearing. Further market imperfections, relative to the idea of a perfect capital market in theory, included bankruptcy risk and the costs of financial distress at high levels of gearing. These reduced and finally reversed the tax shield effect, resulting in an optimal capital structure at the point where the WACC was at its lowest and the value of the company was at its highest.

Other considerations

In practice it has been noticed that companies do not appear to base their financing decisions on the objective of achieving an optimal capital structure, but rather have a preference for sources of finance in the order of retained earnings, bank loans, ordinary debt, convertible debt and equity.

[3]

- iii. Convertible debt is debt that, at the option of the holder, can be converted into ordinary shares. If not converted, it will be redeemed like ordinary or straight debt on maturity. Convertible debt has a number of attractions compared with a bank loan of similar maturity, as follows:
- Self-liquidating - Provided that the conversion terms are pitched correctly and expected share price growth occurs, conversion will be an attractive choice for bond holders as it offers more wealth than redemption. This occurs when the conversion value is greater than the redemption value (if conversion and redemption are on the same date), or when the conversion value is greater than the floor value on the conversion date (if conversion is at an earlier date than the redemption date). If the debt is converted into ordinary shares, it will not need to be redeemed, i.e. self-liquidation has occurred. A bank loan of a similar maturity will need to have all of the capital repaid.
 - Lower interest rate - The interest rate on convertible debt will be lower than the interest rate on ordinary debt such as a bank loan because of the value of the option to convert. The returns on fixed-interest debt will not increase with corporate profitability, so debt providers will have a limited share of the benefits from the investment of the funds they have provided.

When debt has been converted, however, bond holders become shareholders and will potentially have unlimited returns, or at least returns that are higher than the returns on debt finance.

- Increase in debt capacity on conversion - Gearing will increase with a bank loan for the time that the debt is outstanding, and gearing will then return to its previous level when the bank loan has been paid off. Gearing also increases when convertible debt is issued, but if conversion occurs, the gearing will fall not only because the debt has been removed, but will fall even further because equity has replaced the debt. The capacity of the company to service debt (debt capacity) will therefore be enhanced by conversion, compared to redemption of a bank loan of a similar maturity.
- More attractive than ordinary debt - It may be possible to issue convertible debt even when ordinary debt such as a bank loan is not attractive to lenders, since the option to convert offers a little extra that ordinary debt does not. This is the option to convert in the future, which can be attractive to optimists, even when the short- and medium-term economic outlook may be poor.

[3]

2) i. CAPM:-

The relationship between CAPM and the preceding strands of portfolio theory.

- Mean Variance theory assumes that investor's preferences depend only on the mean and variance of payoffs and not on other features. The variance represents risk.
- Portfolio theory suggests if an investor holds a broad range of securities (asset diversification), the individual security risks become less important.
- The Capital Market Line compares risk with reference to a fully diversified investment or market portfolio.
- Under, CAPM investors look at the market risk (or systematic risk) of individual securities (and disregard non-systematic risk which have negligible impact on the portfolio) and how the market risk affects the portfolio risk.
- Market risk is measured by the ratio of covariance of an asset's return with the market return (or beta).
- Formula: $E(r_i) = R_f + \beta_i(E(r_m) - R_f)$

[3]

ii. Advantages of CAPM :-

- Simplicity of use
- Small set of inputs required
- Better than WACC as it takes into account systematic risk

Disadvantages of CAPM :-

- Unrealistic assumptions – perfect market, investors can borrow/ lend at risk-free rate
- Single period transaction horizon
- Ignores other (non-systematic) risks

[3]

3) Passive management v/s Active management

Passive fund management:

- Replicates performance of a benchmark
- No need to 'select' individual constituents except for that required for adjusting relative prices/weightings as per the benchmark.
- Different types of replication: full, partial, synthetic.
- Tracking error and tracking difference can be used to measure the efficiency of replication. Successful replication implies negligible tracking error/difference.

Active fund management:

- Seeks to exceed the benchmark
- Returns should be adjusted for risk and expenses
- Can be top-down or bottom-up OR Growth stock vs Value stock
- Performance measured in terms of 'alpha' return and Sharpe/ Treynor ratios.

Active or Passive fund management for the Indian investment market

The markets are less efficient in an emerging country, like India as compared to a developed country like US. Therefore, where a passive strategy has mostly out-performed elsewhere (as it is harder for fund managers to beat their benchmarks), it is not necessarily the case with India.

Here, active fund management has given historically better returns especially in the long run. In the past 10 years, 60-70% of the funds have outperformed their benchmarks. This can be attributed to the steady growth of the Indian market and professional management of the fund.

There are some sectors of the market which are more suited for passive style. One of India's most efficient segments, the large-cap space (funds with more than 80% allocation to large-cap stocks) has more presence of passive funds. Studies have shown that active management has generated lesser returns for sectors like Technology, Infra and Banking Funds as compared to passive fund management. It should also be noted that there is a lack of passive fund options in India.

There are pros and cons to both strategies. Active funds investment can not only give the investor good returns they can also protect an investor's portfolio from the downside risk if the index goes down. On the other hand, passive funds are cheaper, more liquid and safer. So it is a question of what is more suited for the investor - higher returns (and higher risk) or stable returns (with lesser risk (no fund manager risk)).

Alternatively, we can use a mix of both the strategies. The decision can be broken into two steps:

- Active Asset Allocation
- Passive Component Management

The impact of changing asset allocation is usually much more significant than changing an individual component benchmark, or changing a component between active or passive allocations. An investor, manager, advisor or consultant adds the most value when he gets his asset allocation right. And given the time and effort spent on choosing individual managers, arguably the most efficient way for an investor to spend his time is to focus on actively managing an asset allocation among passively managed components. [8]

4) Three ways of incorporating risk into the investment appraisal process.

- Risk and uncertainty-Risk in investment appraisal refers to the attachment of probabilities to the possible outcomes from an investment project and therefore represents a quantified assessment of the variability of expected returns.
- Sensitivity analysis to changes in project variables – It includes calculating the relative change in a project variable required to make the NPV zero, or the relative change in NPV for a fixed change in a project variable. Only one variable is considered at a time. When the sensitivities for each variable have been calculated, the key or critical variables can be identified. These show where assumptions may need to be checked and where managers could focus their attention in order to increase the likelihood that the project will deliver its calculated benefits.
- Probability analysis - This approach involves assigning probabilities to each outcome of an investment project, or assigning probabilities to different values of project variables. The range of net present values that can result from an investment project is then calculated, together with the joint probability of each outcome. The net present values and their joint probabilities can be used to calculate the mean or average NPV (the expected NPV or ENPV) which would arise if the investment project could be repeated a large number of times.
- Risk-adjusted discount rate - Use the capital asset pricing model (CAPM) to determine a project-specific discount rate that reflected the systematic risk of an investment project.
- Adjusted payback - Payback can be adjusted for risk, if uncertainty is considered to be the same as risk, by shortening the payback period. The logic here is that as uncertainty (risk) increases with the life of the investment project, shortening the payback period for a project that is relatively risky will require it to pay back sooner, putting the focus on cash flows that are more certain (less risky) because they are nearer in time.

[3]

5)

i.

- a) Expected return = $0.5(7.5) + 0.5(12) = 9.75\%$
- b) Correlation coefficient = $27/(11 \times 21) = 0.12$
- c) Variance of returns = $(0.5^2) \times (11^2) + (0.5^2) \times (21^2) + 2 \times 0.5 \times 0.5 \times 27 = 154$
- d) Standard deviation of returns = $\sqrt{154} = 12.4\%$

[2]

ii. Weighted standard deviation = $(11 + 21)/2 = 16\%$

The actual risk, calculated in part (a), is 12.4%. This means that creating the portfolio has led to significant risk reduction. This is not surprising given that the correlation coefficient of 0.12 is low, which means that the returns of the two assets are largely independent of each other. This demonstrates risk reduction through asset diversification.

[1]

- iii. The solution to the measurement of risk for larger portfolios is to regard them as amalgams of pairs of assets. Portfolio risk is the sum of the weighted variances of the individual assets and the weighted covariances.

A critically important feature of portfolio risk emerges as we increase the number of assets in a portfolio. The significance of individual asset risks declines. Portfolio risk is increasingly dependent on the covariance of returns. Individual risk components become progressively less relevant. This is the essence of portfolio theory; diversification makes differences less important. The covariances reflect the shared risks that cannot be neutralised through asset diversification.

The shared risk occurs as multiple covariance calculations. The computational task grows rapidly with the number of assets incorporated into a portfolio, e.g. 10 assets = 45 covariance calculations, 100 assets = 4950 calculations. Furthermore, it complicates the task of constructing portfolios with 'efficient' risk return trade-off due to the growth of possible asset combinations.

[2]

- iv. The portfolio with a correlation coefficient of -0.8 should be preferred. The high level of negative correlation suggests that the two assets' return patterns are highly independent of one another, meaning that a considerable component of the risk of each is offset by the other. By contrast, the portfolio with a correlation coefficient of 0.8 consists of two assets whose returns tend to evolve in very similar ways. This limits the scope of risk reduction.

[1]

6) Indian Bond Market

Recent developments in the Indian Bond Market

- Substantial increase in investments by FIIs and mutual funds due to initiatives taken by Govt & RBI
- Innovations - Basel III bond, Inflation Indexed debenture, 50 year bond, Infrastructure debt bond
- Introduction of Repo in Corporate Bonds (both long-term and short-term securities). Insurers are allowed to participate in these.
- IRDA increased limits to invest in non-AAA rated debt
- Permission for CDS on listed and unlisted rated corporate bonds
- Launch of a separate corporate bond trading platform on NSE for retail investors

Key challenges in the Indian Bond Market

- The bond market in India is dominated by Govt securities. The corporate bond market is still under-developed as compared to other emerging markets
- Limited participation - There is an active market for high rated and big players but dearth at the middle segment. Small and mid-size borrowers are unable to access the debt markets.
- Bulk of the issuances are via private placements, due to less stringent disclosure requirements and lower costs of issue.
- Illiquidity in the secondary market.
- Information asymmetry – Information only available to big players

- High default rates and inadequacy in the legal framework for recoveries in the event of a default.
- Different taxation rules/ tax incentives across the country.

[4]

7) The factors that influence the market value of traded bonds are represented in the bond valuation model.

- Amount of interest payment - The market value of a traded bond will increase as the interest paid on the bond increases, since the reward offered for owning the bond becomes more attractive.
- Frequency of interest payments - If interest payments are more frequent, say every six months rather than every year, then the present value of the interest payments increases and hence so does the market value.
- Redemption value - If a higher value than par is offered on redemption, as is the case with the proposed bond issue of ABC Co, the reward offered for owning the bond increases and hence so does the market value.
- Period to redemption - The market value of traded bonds is affected by the period to redemption, either because the capital payment becomes more distant in time or because the number of interest payments increases.
- Cost of debt - The present value of future interest payments and the future redemption value are heavily influenced by the cost of debt, i.e. the rate of return required by bond investors. This rate of return is influenced by the perceived risk of a company, for example as evidenced by its credit rating. As the cost of debt increases, the market value of traded bonds decreases, and vice versa.
- Convertibility - If traded bonds are convertible into ordinary shares, the market price will be influenced by the likelihood of the future conversion and the expected conversion value, which is dependent on the current share price, the future share price growth rate and the conversion ratio.
- Other factors like tax rate, seniority of debt and whether it is secured or not also influence bond values.

[3]

8) **Features of a bond index:-**

- Should have clear rules for its construction and rebalancing – the composition may change if a bond matures, etc.
- Should be a proper representative – by including a wide range of securities (like short-term, long-term, domestic, international, etc.)
- Should include tradable and liquid bonds and thus it should be possible to invest in the index/ replicate index
- Complete and accurate data
- Full disclosure and transparency in calculation
- Low index's maintenance and transaction cost

[3]

[Total Marks-47]

Solution 2 :-

- i. Events that trigger payments under credit derivatives are known as credit events. Examples of credit events include:
- bankruptcy (insolvency, winding-up, appointment of a receiver)
 - a rating downgrade
 - repudiation – when the debt issuer simply chooses to cancel all of the outstanding interest payments and the capital repayment of the debt
 - failure to pay a particular coupon
 - cross-default - where a credit event on another security of the issuing firm will also be considered as a credit event on the bond in question
 - debt restructuring – the configuration of the debt obligations is changed in such a way that the credit holder is unfavorably affected
 - obligation acceleration – the debt obligations of the issuer become due before their originally scheduled maturity date

[2]

- ii. CDS is an OTC credit derivative contract between two counterparties where the risk of default (or credit risk) is transferred from the holder of the debt security to the seller of the swap.

The “protection buyer” purchases a CDS from the “protection seller” by making periodic payments (typically either semiannually or quarterly), called a CDS premium, to the “protection seller”, and in return receives a payoff if an underlying financial instrument defaults or experiences a similar credit event.

If the credit event does not occur within the term of the contract, the buyer receives no monetary payment but has benefited from the protection during the tenure of the contract.

There are two settlement options.

Cash settlement – where a pure cash payment, representing the fall in the market price of the defaulted security is paid by the protection seller.

Physical settlement – which includes the exchange of both cash and the security. The protection seller pays the buyer the full notional amount and receives, in return, the defaulted security.

[2]

- iii. Arbitrage based pricing is based on the idea that if two assets have the same payoffs in all future states of the world, they ought to have the same price in a perfect financial market. In theory, if the prices of these two assets diverge, market participants will short sell the more expensive asset and purchase the less expensive one, and as a result make a sure profit without making any investment, and without taking any risk.

Suppose that the arbitrageur buys a corporate bond with a maturity of T years and also buys default insurance via the purchase of an equivalent maturity CDS contract at a price of p (percentage of nominal). If the yield to maturity of the bond is y , the annual return to the arbitrageur is equal to $y - p$. This return should approximately equal the T year risk-free rate (denoted r) because the resulting investment is now insured against default.

Thus, if $y - p > r$, the arbitrageur should theoretically enter into the following transactions to exploit the mispricing:

- Purchase T year bond with yield y and price B .
- Purchase CDS for maturity of T years at p .
- Borrow B at the risk-free rate to finance the purchase.

On the other hand if $y - p < r$, the transactions correspondingly involve:

- Sell T year bond with yield y and price B
- Sell CDS for maturity of T years at p
- Lend B at the risk-free rate and receive as collateral the bond that is sold

Thus, in both cases, the arbitrageur makes risk-free profit without any investment.

For market equilibrium to hold true, $y - p = r$.

[4]

iv.

CDS can be used for

- Hedging, transferring or mitigating credit risks
- Managing regulatory and economic capital requirements
- Speculation, trading & market making
- Active portfolio/ asset management
- Protection sellers can enhance their yields and diversify credit exposure
- Gaining additional exposure on a particular counterparty in excess of exposure norms

Constraints on the use of CDS

- Do not work under economic downturn
- Regulatory restrictions specially on who can write or sell the CDS protection
- Inadequate systems and infrastructure
- Negative mindset and lack of knowledge of the product
- Lack of statistics/ past data which leads to unreliable pricing
- Mostly tailor-made. Lack of standardization of contracts

[4]

v. **The various risks in CDS are as given below:**

- a) Counterparty risk – Arises in a credit event when the asset values fall down and the seller of the protection against the credit default does not have the requisite arrangements to make good the losses.
- b) Concentration risk – Arises if the protection seller has sold to too many parties leading to large open positions sometimes more than the value of the underlying itself. It can lead to systemic risks.
- c) Basis risk- Arises when CDS providing exactly matching protection may not be available. CDS covering the exact remaining maturity of the bond may not be available.

- d) Operational risks – Trade related processes, outstanding trades, valuation and settlement of the contracts
- e) Legal risks – Regulatory risks, non-compliance etc

[3]

vi. Risk management framework for CDS

- Carefully evaluate the type of assets on which protection can be bought
- Best to buy protection from exchanges if available otherwise do a check on which Counterparties from whom CDS can be bought and limits on the counterparties
- While buying the CDS protection check the amount of protection already sold by the counterparty and also whether there is any limit up to such protection will be sold
- Valuation & reporting norms
- Stress testing on the capability of the counterparty to meet its obligations at periodic intervals
- Margins related rules and restrictions – what can be held as margin, etc.
- Settlement of MTM obligations
- Necessary systems and controls as prescribed by the Regulator
- Periodic review of the policy

[3]

[Total Marks-18]

Solution 3 :

i. Macro-economic situation

The macro-economic situation in India has to be viewed in the context of global macro-economic environment.

The global economy grew by around 3% in 2013 and continued its recovery in 2014 from the global financial crisis of 2007-08. However the following pose downside risk to this recovery:

- the ongoing tapering of quantitative easing in the US
- the weak balance sheet in the EURO area
- the inflationary pressures in the emerging market

The growth of Indian economy weakened over the past couple of years falling below 5% due to a sharp fall in Industrial growth to below 1%. The future prospects depend on a range of global and domestic factors, the outcome of the general elections being one of the key domestic factors.

The annual average CPI has touched double digits or stayed just below for last six years. The Reserve Bank of India continues to raise benchmark interest rates to bring inflation under control.

India's current account has remained in deficit for close to a decade due to large trade deficits. The Government (Centre and State combined) have been running fiscal deficits of around 7% of GDP.

The Government debt to GDP ratio, however, has fallen over the past few years as the debt grew at a pace slower than the nominal GDP.

The Rupee depreciated by over 14% against the US dollar in 2013. The tapering of QE, widening current account deficit, and slowdown in the Indian economy are some of the widely considered reasons for this depreciation.

[5]

ii. Investment avenues available to foreign investors for making investment in India

- a. Foreign Direct Investment (or FDI)** – While used primarily by companies to expand operations, the FDI route is also used by individual investors looking to buy unlisted investments, either directly or through private equity funds, as “portfolio investments”.

Certain industry sectors have restrictions on FDI. Investors may perform FDI via the “automatic” route if they wish to invest in certain approved sectors. Under the automatic route, there is no requirement for regulatory approval. If the investment amount exceeds the specified sector cap, or if it is in a non-approved sector, investment may be made under the “prior approval” route. This requires government approval.

- b. Foreign Institutional Investor (or FII)** – Non-Indian institutions can invest in India under the “Portfolio Investment Scheme” by registering with SEBI as an FII. Both end investors (asset owners such as pension funds, sovereign wealth funds, endowments or banks) and intermediary institutions can use this route.

Although the Portfolio Investment Scheme was not designed for individual investors, FIIs can offer them “sub-accounts”.

- c. Qualified Foreign Investors (or QFI)** – In 2011, India introduced the QFI route, designed to encourage foreign individual investors to invest directly into Indian domestic mutual funds or the equity and debt markets, without going through an FII or registering as an FII sub-account holder.

The QFI route gives high net worth individuals the opportunity to invest in Indian debt, which offers attractively-high interest rates and returns. Previously, such investors were only able to invest in non-convertible debentures listed on the stock exchange (via the FII route).

- d. American depositary receipt (ADR) or global depositary receipt (GDR)** - It is a way for foreign investors to invest in companies whose shares are listed in India just like normal shares. The holder of an ADR or GDR is entitled to all benefits such as dividends and rights issues from the underlying shares. They are sometimes – but not always – able to vote. Unlisted companies that have not yet accessed this route for raising capital need to be listed in the domestic market either prior or simultaneously to this.

- e. Foreign Currency Exchangeable Bond or FCEB** is

- A bond expressed in foreign currency.
- The principal and the interest of which is payable in foreign currency.
- The issuer of the bond is an Indian company.
- The bonds are subscribed by a person resident outside India.

- The bonds are exchangeable into equity shares of another company which is also called the offered company.

Foreign currency convertible bonds (FCCBs) are issued by a company to non-residents giving them the option to convert them into shares of the same company at a predetermined price. On the other hand, foreign currency exchangeable bonds are issued by the investment or holding company of a group to non-residents which are exchangeable for the shares of the specified group company at a predetermined price.

[10]

iii.

a. The split between capital (7 million) and loans (8 million) looks reasonable given that the client is a young professional whereby her income is expected to grow and she will have sufficient capacity to repay her loans in future. The size of the loan does not seem to be very large given her current income levels.

On the different loans and whether it should be continued or not let us evaluate each loan separately:

Education Loan @ 14% pa:

Key features

- The interest payable on education loan is deducted from the income to calculate taxable income if the loan meets the specified qualifying criteria. Needs to check whether the loan has such advantage or not.
- The interest rate on the loan is usually a few percentage points (2-5%) higher than the return that can be generated from safe fixed interest investments, like bank fixed deposits.
- The loans usually have a moratorium period during which the loan accumulates and no repayment is required.

What the client could do?

- The client has sufficient liquid funds to foreclose the loan.
- The client will have to give due consideration to any foreclosure conditions especially foreclosure charges and penalties before choosing to foreclose.
- The tax benefit from payment of interest on the loan could outweigh the cost of excess interest paid on the loan. It may therefore be efficient to not foreclose the loan until such tax benefits are available.
- The client could re-negotiate the terms of the loan with the lender to reduce the interest burden.

Housing loan: This one seems to be a good deal amongst all the loans she has availed since it provides an opportunity to put money in the loan account and reduce the interest outgo thereby generating 12% on such liquid money which can be withdrawn whenever required. This one should not be foreclosed soon since it works like an OD account which is normally available to corporates and not to Individuals. In fact she can put her cash and bank balances in this account and generate 12% return on the same instead of the lower returns of 4-5% (which a bank savings account provide) or nil returns (on idle cash).

Even on the remaining outstanding loan the interest which is paid is generally tax deductible thus the actual interest outgo on this loan might be in the range of 8% per annum.

Car loan @ 13% p.a.:

- The key difference between car and education loan is the availability of tax benefit. There is usually no tax benefit available on car loans.
- Other features and considerations are very similar to the education loan.
- The client has sufficient funds to foreclose the loan and in the absence of any tax benefit she should consider foreclosing it since the interest cost is higher than what she can generate on her cash/bank deposits.

[4]

b. The key features of different assets are given below:

House:

- This is a real return generating asset
- A good hedge against inflation since price of flats have been growing at inflation rates or better than inflation.
- We are not sure but if the client is utilising it for her personal residence then it saves the cost of renting a flat as well.
- Has a touch and feel effect and may give a psychological comfort of owning your own home
- Has disadvantage in terms of liquidity aspect, cannot be sold off easily
- Transaction costs on sale like taxes (in case it is sold within 3 years of possession) and brokerage charges paid are high.

Two Cars (1.2 mn):

Key features:

- they are depreciable and usually no-revenue generating assets
- It will be compulsory for the client to buy motor insurance on both the cars
- The key consideration with car as an asset is to
 - minimize the depreciation (maximize the resale value) through good driving habits and regular maintenance
 - minimize insurance premium by maintaining a good driving record and negotiating with insurance providers

Provident fund (1.8 mn)

Key features of this asset:

- the contributions to EPF are mandatory for salaried individuals unless exempted under certain conditions
- the contributions made by employees are tax deductible under section 80C of the Income Tax Act

- the employers contributions are subject to admin charges and compulsory contribution to employees deposit linked insurance
- the interest rates on provident fund balance are declared at the end of each year – the returns are not known in advance
- the interest credits have been below inflation for corresponding periods in the recent past implying a negative real return
- the interest earned is exempt from tax
- the returns and withdrawal (after the lock-in period) are tax exempt
- withdrawals/loans from the fund are possible under certain conditions but members could experience practical difficulties in getting their withdrawal requests processed in a timely way

What the client will have to do?

- since the contributions are mandatory for salaried individuals, the client will have to continue to contribute to the fund
- the returns are comparable to other safe fixed interest investment instruments like bank fixed deposits
- The tax deductible nature makes it more attractive compared to fixed deposits and the client should utilize this benefit to the full extent available in each financial year.

Mutual Fund (Equity):

This is a good investment given that the client is still on her thirties and has a steady flow of salary. This may be increased gradually through systematic investment plans (SIPs) wherein entry is made in the fund every month and not exposed to price volatility which may be seen if all the money is invested in one lump-sum.

As the age grows the client may decide to put money into balanced fund instead of equity fund since her risk taking abilities may come down and as given she has a balanced approach to risk even now.

Insurance:

The future income earning capacity is very high given that the current salary levels are good and she still has age on her side wherein she can work for 20 years or more. Hence she requires a large protection cover which is not there. This will work as a hedge against the unfortunate event of death. She also requires a hedge against medical emergencies and thus a health insurance. This will also help by releasing a large amount of money currently being held in liquid assets (cash and bank balances).

Cash (7.0 mn)

- Cash is a very liquid asset, access to it is almost instantaneous with the availability of debit cards and online access to funds
- Cash with reputed banks in India are perhaps the safest financial instruments
- the returns earned are likely to be very low compared to fixed interest investments (could vary between financial institutions)
- Cash is a good asset for emergency funds and intermediate funds waiting to be deployed

[4]

c. How the client can restructure her assets and liabilities.

- The client should keep between, say, 5-10% of total assets in cash (or other liquid assets).
- The liquid money (cash and bank balance) should be parked in the OD account of home loan wherein the loan amount outstanding would reduce. In case of any emergency cash can be withdrawn from that account and used. This will help her earn higher return on her liquid money as compared to normal bank savings account returns.
- She should buy term insurance and critical illness covers to give adequate protection to her dependents, if any, against loss of income which may happen in case of her untimely death or critical illness.
- Health insurance to hedge against sudden liquidity requirements which may arise in case of medical emergency
- Some cash may go towards repayment of outstanding car loan since it does not have any tax advantages and is expensive at 13%. This would increase her annual saving to the extent of the monthly installment of such loan.
- Even some amount of education loan may be repaid but should be evaluated against the tax advantages available on such loans.
- All loan repayments need to be compared with the opportunity of parking money in the OD account wherein there is a flexibility of withdrawal and the return generated on the amount put into this account will be 12% per annum (given the loan interest rate is 12% p.a.) adjusted for tax benefits.
- Car is a depreciating asset so if 2 cars are not required then it is better to dispose of one and repay the loans against it.
- Mutual fund investment may be increased through SIP route since it is aligned to her regular monthly income through salary.

[7]

iv. The unlisted company is owned by a large listed conglomerate.

The process of constructing synthetic shares begins with carefully evaluating the conglomerates' constituent parts and use short position to remove all that it owns other than the soon-to-be-listed insurance company.

The large conglomerate could have holdings in lots of smaller listed companies other than its core businesses. The investor will have to identify traded market index/indices that closely match the conglomerates' core businesses.

The investor could then construct the following portfolio:

- long position in the conglomerates' shares
- short position in traded market index/indices representing the core business done by the conglomerate or if such indices are not available then take short position in other listed companies who have similar / matching core business as this conglomerate
- short position in the listed constituents of the conglomerate

The net exposure thus created would be that of the soon to be listed company.

There could be significant basis risk in such a synthetic construction especially if there other significant non-listed holdings in the portfolio or traded indices for conglomerates core business are not available.

[5]

[Total Marks-35]
